

DISHANTKUMAR SUTARIYA

Bergerstr.7, 28217, Bremen, Germany

+49 162 476 9215

dishantsutariya24@gmail.com

linkedin.com/in/dishant-sutariya

github.com/dishant24

dishant-sutariya/portfolio



SUMMARY

AI Researcher specializing in medical imaging and fairness with 2 years of experience building production-grade AI models using Python, Pytorch, and MLOps tools (Docker, AWS, MLflow). Experience in end-to-end model development, model evaluation, and deploying AI for clinical applications. Seeking roles where technical innovation directly translates to real-world patient outcomes.

EXPERIENCE

Fraunhofer Institute for Digital Medicine MEVIS Researcher

Oct 2025 – Mar 2026
Bremen, DE

Chest X-ray Fairness and Bias Mitigation

- Developed and validated a CheXmask lung cropping preprocessing pipeline to mitigate racial bias, reducing race-prediction AUROC from 0.623 to 0.593 while maintaining diagnostic AUC at 0.738–0.84 across diverse patient demographics
- Achieved 0.84 AUC on chest disease classification using CLAHE normalization and Isotonic Regression calibration, matching DenseNet-121 baseline while improving fairness and deployment robustness
- Lead author for BVM 2026 accepted paper on preprocessing methods for racial encoding, defining new protocols for robust and fair chest X-ray diagnosis

Working Student

Dec 2024 – Sep 2025
Bremen, DE

meval Statistical Toolbox Development

- Engineered meval Python library (Plotly-based) for statistical model evaluation and fairness validation, reducing manual bias review time by 20% (approximately 8 hours per week) for research teams
- Co-authored MICCAI 2025 conference paper “meval: A Statistical Toolbox for Fine-Grained Model Performance Analysis”; achieved first-submission acceptance

Tech Stack: Python, Pytorch, Plotly, Statistical Analysis, Segmentation, Histogram Matching, Model Evaluation

Maruti Techlabs Machine Learning Intern

Aug 2021 – Jul 2022
Gujarat, IN

Vehicle Insurance Prediction

- Developed vehicle insurance claim prediction model achieving 92% accuracy (12% improvement over baseline) across 50,000 customer records by identifying key features (vehicle damage, prior claims history, driving experience)
- Implemented SMOTE to resolve class imbalance, improving F1-score by 10% and reducing false negatives by 8%, directly minimizing financial risk
- Benchmarked XGBoost vs. Random Forest, deployed optimal XGBoost model with 10-fold cross-validation for production stability

Tech Stack: Python, XGBoost, Scikit-learn, Pandas

EDUCATION

University of Erlangen-Nürnberg

Oct 2022 – Sep 2025
Erlangen, DE

Master of Science – Medical Image and Data Processing

- Final Grade: 2.2 / 5.0

Gujarat Technological University

Aug 2017 – Jul 2021
Gujarat, IN

Bachelor of Science – Computer Engineering

- Final Grade: 1.7 / 5.0

SKILLS

- Programming:** Python (expert), SQL (intermediate)
- ML & DL:** Scikit-learn, Pandas, Pytorch, Tensorflow, Opencv, RAG, LangChain, HuggingFace, Transformers
- DevOps:** Docker (intermediate), MLflow, Weights & Biases, AWS (EC2, S3, SageMaker), GitHub Actions (intermediate), Git, CI/CD
- Others:** Microsoft 365 Office, Power BI, VS Code, Jupyter Notebook, HL7 & FHIR
- Languages:** English (B2), German (B1), Hindi (Native), Gujarati (Mother Tongue)

PUBLICATIONS

- Sutariya, D., Petersen, E. (2026). meval: A Statistical Toolbox for Fine-Grained Model Performance Analysis. *Fairness of AI in Medical Imaging* (Lecture Notes in Computer Science, vol 15976). Springer, Cham. DOI: 10.1007/978-3-032-05870-6_19
- Designed statistical toolkit for analyzing AI model performance metrics and fairness in medical imaging applications
 - Addressed critical issues of bias detection and mitigation in medical AI systems
- Sutariya, D., Petersen, E. (2026). Impact of Preprocessing Methods on Racial Encoding and Model Robustness in CXR Diagnosis. In: Handels, H., et al. *Bildverarbeitung für die Medizin 2026. BVM 2026. Informatik aktuell*. Springer Vieweg, Wiesbaden. DOI: 10.1007/978-3-658-51100-5_29
- Extended thesis work implementing lung cropping as preprocessing method on MIMIC-CXR and CheXpert datasets
 - Experiments repeated across five random seeds to ensure robustness and reproducibility
 - Paper accepted for publication; poster presentation in preparation

AWARDS AND ACHIEVEMENTS

- Best Poster Presentation Award**

Workshop Fairness of AI in Medical Imaging (FAIMI) at MICCAI 2025

 - Awarded for poster: “meval: A Statistical Toolbox for Fine-Grained Model Performance Analysis”
 - Authors: Dishantkumar Sutariya, Eike Petersen
- First Place – Design and Analytics Summer Challenge**

Siemens Healthineers

 - Participated in 2-day hackathon focused on chest X-ray image registration to standard lung template for lesion distribution monitoring
 - Developed and presented solution as part of team; secured first place through democratic voting
 - Gained hands-on skills in image registration and medical data analysis under industry expert guidance
- Sep 2025

Busan, South Korea
- Sep 2023

Forchheim, DE

PROJECTS

- University of Erlangen-Nürnberg**

Diffusion Model for Energy-Level Transformation in Photon Counting CT

 - Engineered Vision Transformer (ViT) conditioned Diffusion Model in PyTorch to synthesize high-quality photon counting CT images at different energy levels from 50 keV baseline
 - Achieved Dice Score of 0.86 through optimization of multi-objective loss function (MSE, SSIM, Perceptual Loss)
 - Maximized structural integrity and material discrimination for energy-resolved medical imaging applications

Tech Stack: Python, Pytorch, Tensorflow, Diffusion Models, ViT, and U-Net
- Personal Project**

Simple RAG Q&A Application – Retrieval-Augmented Generation System

 - Designed and developed RAG (Retrieval-Augmented Generation) question-answering system enabling users to query local documents (PDF, TXT, CSV, Excel, Word, JSON) with LLM-powered answers based on document context
 - Implemented FAISS vector store for efficient semantic search across multi-format document embeddings, enabling fast information retrieval from large document collections
 - Integrated Groq LLM API for fast inference, combining vector-based retrieval with generative AI to provide accurate, context-grounded responses
 - Repository: github.com/dishant24/Simple_RAG_Application

Tech Stack: Python, FAISS, LLM API, Langchain, HuggingFace, Vector Embeddings, RAG Pipeline
- Mar 2024 – Oct 2024

Erlangen, DE
- 2026 – Present

GitHub